EIGSEP Memo 002

Marjum Pass 2026-07: images, camera poses and positions

EIGSEP Team

Living document. This version: 2026-10-10

Abstract

Where the antenna and transmitter were, how we know, and how well. The positions come from photogrammetry of 37 field photographs against a digital elevation model (DEM); the absolute azimuth reference comes from GPS points on the highline anchors. The adopted geometry is `v0004`: a joint fit of camera poses, landmarks, antenna and transmitter in the EPSG:6341 UTM raster grid of DEM `v0002`. It is a point fit selected by exact target density from non-converged MCMC chains, so it has no posterior width. Refits that withhold one kind of evidence at a time bound the antenna to 3.2 m and the transmitter to 2.7 m (hard brackets set by the worst independent check). This memo states the adopted positions, those checks, how they compare with the historical release `v0001`, what the joint MCMC does and does not establish, and the camera and coordinate conventions needed to reuse any of it. Every figure and table is generated by `make_products.py` in this directory (Appendix C). The observing configuration and angle conventions are in memo 001.

1 Summary

- **Adopted positions** (geometry `v0004`, UTM raster grid, metres; Table 1): box-air in the 91 m era at (1656.22, 2031.83, 1778.60), UTM (292656.5, 4347032.1), 94.1 m above the DEM surface; the transmitter at (1652.18, 2025.33, 1683.48), UTM (292652.4, 4347025.6). **Bounds: 3.2 m (antenna) and 2.7 m (transmitter)**, set by the worst held-out check (§4.1); they are brackets, not posterior sigmas.
- **In-sample closure**: at the v0004 poses the 23 labelled antenna rays miss the antenna by a median of 0.74 m and at most 2.93 m (IMG_2235); 10 miss by more than 1 m. The 8 transmitter rays miss by at most 0.42 m. Every pose was fitted with these labels, so these residuals are not independent checks.
- **Held-out checks** (§4.1): leaving each antenna label out in turn, its ray misses the refitted antenna by a median of 1.04 m and at most 3.18 m (IMG_2232). Without their own transmitter labels, the two unconditioned views IMG_2210/2211 miss the transmitter by 1.9–2.6 m. Withholding all HEIC GPS moves the antenna 0.21 m and the transmitter 0.15 m: the DEM, not the GPS, places the network.
- **Relation to the historical release**: v0004 lies 1.5 m (antenna) and 0.9 m (transmitter) from the `v0001` positions, mostly in height.
- **The joint MCMC did not converge**: 212 of 213 global coordinates fail $\hat{R} \leq 1.01$; 211 fail bulk ESS ≥ 400 and 200 fail tail ESS ≥ 400.
- **Azimuth zero** is $37.84° \pm 0.6°$ true from the GPS highline anchors. The LIDAR terrain fit at the v0004 antenna, with bearings now applied on the UTM grid, gives $36.27° \pm 0.45°$, 2.1σ away; the historical fit gave 1.4σ, and the frame correction, not the new antenna, raises it.
- **Transmitter orientation**: from the feed's corners in IMG_2203 at the v0004 poses, the rx6 dipole axis lies at 165.5° true (mod 180; NNW–SSE) and rx1 at 76°. The earlier 167.3° was a grid bearing. A theodolite-app reading lies 1.2° away (§9).
- **Open**: poses that depend on their own labels (§4.1); the 2.1σ LIDAR–GPS az-zero tension; and the tether-anchor position (needed for the field heights).

2 The images

37 photographs are archived in `marjum-2026-07/imgs/` (HEIC, iPhone, EXIF times in MDT; Appendix A). They fall in two groups:

- **6 transmitter-era frames** (IMG_2159, 2171, 2172 on 07-15; IMG_2198, 2199, 2203 on the morning of 07-17), taken near the transmitter with the transmitter labelled. In the historical release their poses came from a joint fit of terrain horizon, image features and the transmitter, so they are *transmitter-conditioned*: their own transmitter rays are not independent evidence for its position. In the v0004 model inputs only IMG_2159 and IMG_2199 carry tie points (88 each), and those link only each other.
- **The 91 m parallax set**, IMG_2209–2245 (07-17 14:42–15:10 MDT), taken on a hike up the west slope with the antenna at its final height and labelled in each frame. In the historical release their poses came from earlier terrain-horizon and feature fits and were held fixed, which made them independent evidence for the antenna. IMG_2210 and 2211 also see the transmitter.

In `v0004` all 29 usable poses are sampled jointly with the landmarks, the antenna and the transmitter, so no camera is independent of the labels any more. 29 frames have usable poses. 8 of the parallax set are excluded: seven because horizon registration settled in a false terrain minimum, one (IMG_2219) for borderline registration. Two further photographs are in neither geometry: IMG_2202, taken at the transmitter looking across the feed, with its own horizon fit (§9); and IMG_2204, a theodolite-app capture looking down on the transmitter.

3 Camera model and conventions

Frame. Positions are E, N, U in metres on the UTM raster grid of `derived/dem/v0002/marjum_dem.npz`: EPSG:6341 easting and northing minus the southwest pixel-centre origin (291000.25, 4345000.25) m, and U on the DEM's vertical datum. The DEM is USGS 3DEP elevation on a 0.5 m grid, stored as float32. UTM grid north differs from true north by 1.52° at the highline. GPS positions enter this grid with a provisional 3 m vertical survey offset and no horizontal offset; the offset has not been checked against independent survey features. The historical release `v0001` used the same elevation pixels (`derived/dem/v0001`) at the same raster coordinates, but converted GPS positions through a tangent plane with a $(-11,\ 36,\ 3)$ m survey offset. Its camera, antenna and transmitter numbers are therefore comparable with v0004 relative to the terrain, but its GPS-referenced quantities (anchor stations, GPS residuals) are not.

Camera. Each pose is (E, N, U, θ, ϕ, tilt, f) with $R_{\text{body}\to\text{ENU}} = R_z(\phi)\,R_y(\theta)\,R_z(\text{tilt})$. The ray through pixel (x, y) is, before rotation, $[(c_y - y)/f,\ (c_x - x)/f,\ 1]$ with (c_x, c_y) the image centre, after removing radial distortion $q_d = q\,(1 + k_1 r^2 + k_2 r^4)$ in focal-normalized coordinates. The 91 m parallax cameras share (k_1, k_2) within each of two lens groups; each transmitter-era camera has its own. Distortion is held fixed in `v0004`. **Pixel rows count from the bottom**: the fitting code loads each photograph upright and then flips it vertically (as recorded in the release), so a label or projection at row y plots at row $H - y$ on the upright image.

What constrains a pose. Two kinds of data tie a camera to the terrain. *Tie points* are image features matched between photographs of the same scene (Fig. 1); each becomes a 3-D landmark whose position is fitted along with the poses, and every observation of it contributes the pixel distance between where it was seen and where the landmark projects. The 91 m parallax cameras carry nearly all of them; of the transmitter-era frames only IMG_2159 and IMG_2199 have any, and only to each other, which is why the pair shown is IMG_2222/2223. The *horizon* constrains every camera on its own: the DEM, ray-traced through a trial pose, must put terrain where the photograph's segmentation finds ground and sky where it finds sky (Fig. 2). Figure 3 shows both checks at the fitted poses for the tie-point pair at the v0004 poses: the model horizon follows the segmented one, and the v0004 landmarks reproject onto their observed pixels with median residuals of 2.7 and 2.4 px, within the model's 4 px tie-point scale. Appendix B shows the horizon check for all 29 usable images: the median horizon offset is 9 px, and the worst is 19 px (IMG_2231).

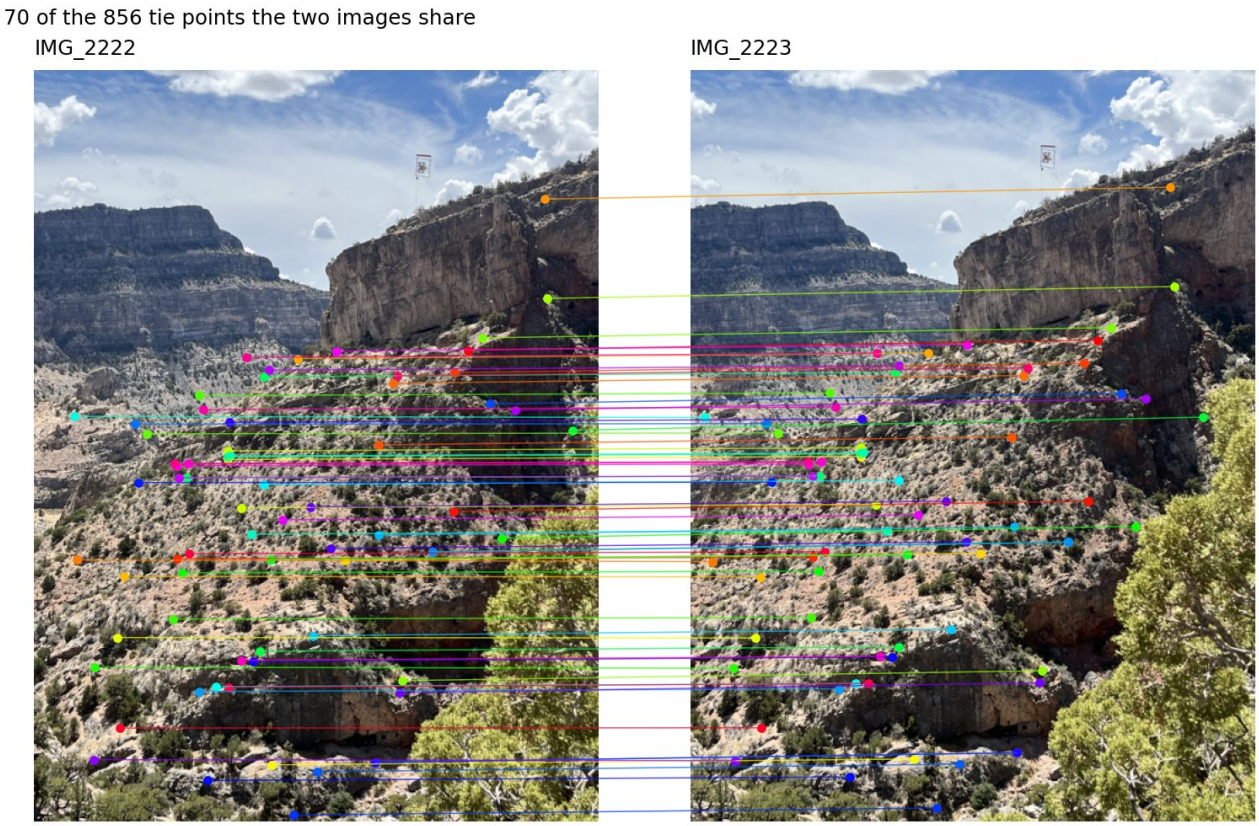


Figure 1: Tie points: a random 70 of the 856 landmarks observed in both IMG_2222 and IMG_2223, two frames from neighbouring positions on the 91 m traverse, each joined to its match in the other image.

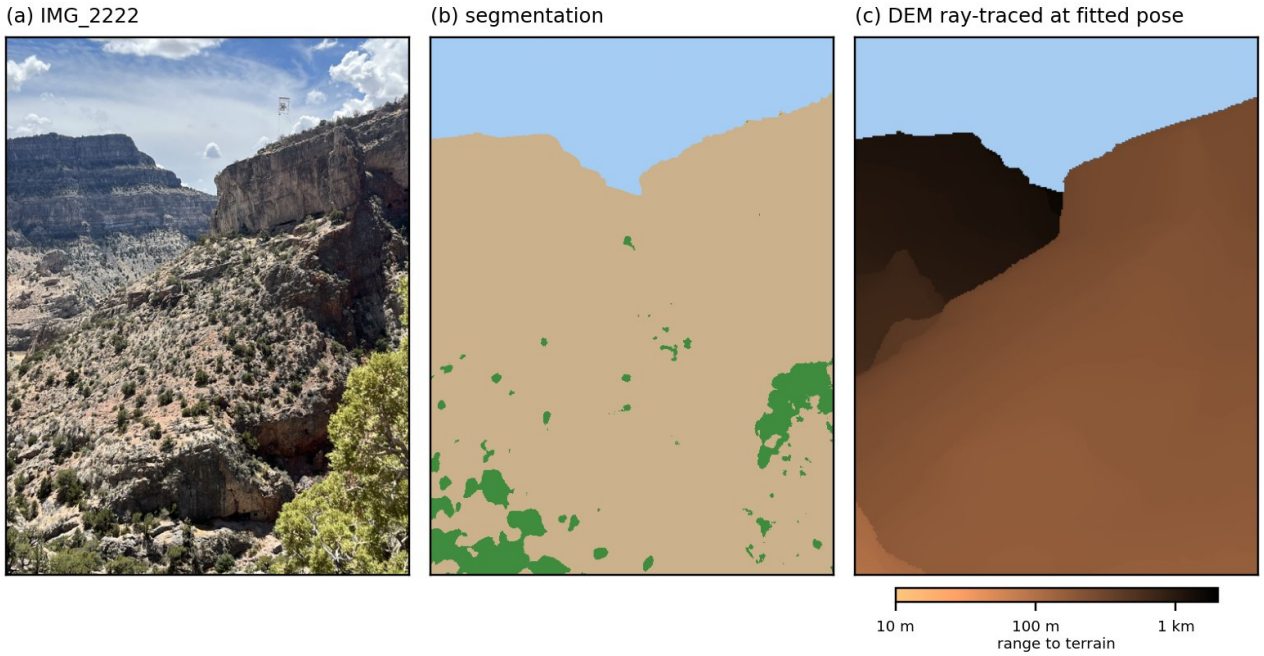


Figure 2: The horizon constraint for IMG_2222, the left frame of Figs. 1 and 3. (a) The photograph. (b) Its segmentation (`derived/geometry_memo_inputs/v0001/img_seg_IMG_2222.npz`): sky, ground, trees. (c) The DEM ray-traced through the v0004 pose: each pixel's ray (on a 12 px grid) is marched in 2 m steps to its first terrain hit, coloured by range; rays that reach 2.5 km unobstructed are sky. Trees, which the DEM does not contain, are why the segmentation marks them separately.

4 Antenna and transmitter positions

Geometry `v0004` (`derived/geometry_posterior/v0004`) is the adopted release; consumers read it as `imgs/fits/v0004_marjum_geometry`, whose `shared.json` carries these positions and their UTM values. Table 1 gives its positions; §6 describes the run and why this state was selected. Figure 4 shows the usable cameras at their v0004 poses, the ray from each through its labelled antenna or transmitter pixel, and how far each ray passes from the v0004 position.

Table 1: Adopted positions (geometry `v0004`). Grid coordinates are the UTM raster grid of DEM v0002 (§3); UTM is EPSG:6341. "Above DEM" is U minus the DEM surface at (E, N). Ray misses are at the v0004 poses. "v0004 − v0001" is the explicit comparison with the historical release. The last two rows compare the v0004 chain means (pooled over chains) with the selected v0004 state, and give the spread of the chain means (§5); the chains did not converge, so neither is a posterior width. The bounds (3.2 m, 2.7 m) are in §4.1.

	antenna, 91 m era	transmitter
grid E, N, U [m]	(1656.22, 2031.83, 1778.60)	(1652.18, 2025.33, 1683.48)
UTM E, N [m]	(292656.5, 4347032.1)	(292652.4, 4347025.6)
above DEM [m]	94.1	0.3
labelled rays	23	8
ray miss, median / max [m]	0.74 / 2.93	0.06 / 0.42
v0004 − v0001, E, N, U [m]	(+0.36, +0.91, +1.20)	(-0.02, +0.22, -0.91)
v0004 chain mean − selected [m]	(-0.24, -0.25, +0.08)	(-0.03, +0.44, +0.25)
between-chain sd of means [m]	(0.20, 0.11, 0.14)	(0.75, 1.07, 0.28)

Uncertainty. The v0004 chains did not converge (§6), so their spread is not a posterior width. The bounds instead follow the rule the historical `v0001` release used: each is the worst independent check, rounded up to 0.1 m, a hard bracket rather than a sigma. For the antenna that is the worst held-out ray, 3.2 m; for the



Figure 3: IMG_2222 and 2223 at their fitted poses. Green: segmented sky boundary; red: DEM horizon (maximum elevation angle over 5 m–3 km per azimuth) projected through the pose. Circles: observed tie-point pixels (a random 120 of the shared landmarks); crosses: the fitted landmark positions projected through the pose. Residuals are a few pixels; the insets show a 120 px patch at full resolution. Poses and landmark positions are geometry `v0004`.

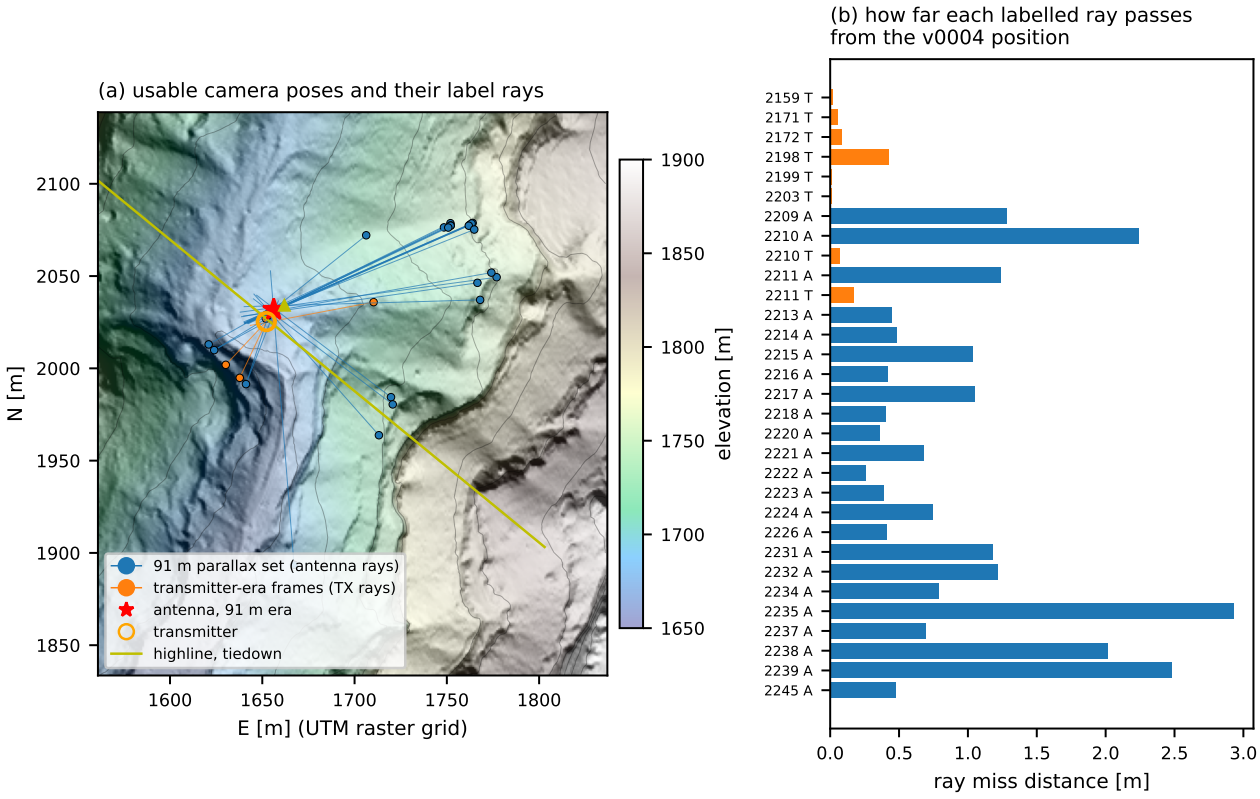


Figure 4: (a) Usable camera poses (geometry `v0004`) on the DEM hillshade, coloured by elevation, with the ray through each labelled antenna (blue) or transmitter (orange) pixel, drawn to just past the target. The highline runs between its GPS anchors (yellow); the triangle is the pulley-plate tiedown. (b) The perpendicular distance from each labelled ray to the v0004 position (A: antenna, T: transmitter). Every pose was fitted with these labels, so the distances are in-sample residuals.

transmitter, the worst miss of an unconditioned view refitted without its own label, 2.7 m (§4.1). The historical v0001 brackets (1.7 m and 1.5 m) were set on fixed release poses and do not test v0004.

Antenna. At the v0004 poses the antenna rays miss by a median of 0.74 m and at most 2.93 m (IMG_2235); 10 of 23 miss by more than 1 m. On the v0001 release poses the same rays missed the v0001 antenna by a median of 0.32 m and at most 1.61 m. The v0004 model carries a fitted extra scatter on the antenna labels, 27 px at the selected state against the 3 px label scale, so the joint fit trades antenna-label agreement for horizon, tie-point and GPS terms. Whether that reflects the labels, the antenna moving during the hike, or pose error is untested. Only the 91 m era has frames with the antenna labelled, so this position applies to that era alone; nothing establishes that the antenna kept its horizontal position across lifts.

Transmitter. The 8 transmitter rays miss by a median of 0.06 m and at most 0.42 m (IMG_2198), with a fitted extra label scatter of 5.7 px. In the historical release the two unconditioned cameras IMG_2210/2211 placed the transmitter about 1 m higher than the transmitter-era cameras did. At the v0004 poses, which were fitted with those labels, their rays pass 0.10 m apart and within 0.12 m of the v0004 transmitter; §4.1 tests them without their labels. The transmitter is 95.4 m from box-air (95.1 m of it vertical), 4.61° from nadir, and 0.3 m above the DEM surface.

4.1 Held-out checks

Each check removes one kind of evidence from the v0004 model, refits only what that evidence touched, and scores the removed evidence (Table 2; `derived/geometry_checks/v0001`). The refits maximize the model's exact log density, starting from the v0004 state; the chains did not converge, so there is no posterior to marginalize. Everything not refitted stays at v0004, which is the scope of each result: in particular, the other cameras' poses and the landmarks were fitted with all labels.

Table 2: Held-out checks of geometry `v0004`. Each row withholds one kind of evidence, refits what it touched, and scores what was withheld.

evidence withheld	refitted	result
Antenna, leave one label out	23 refits	held-out ray miss median 1.04, max 3.18 m (IMG_2232) antenna jackknife s.e. (0.58, 0.36, 0.26) m (E, N, U)
Transmitter, without IMG_2210/2211 labels	8 poses + transmitter	six-camera transmitter moves 0.02 m; the pair's rays miss it by 1.88–2.61 m
Network, without HEIC GPS	translation + 29 poses	antenna moves 0.21 m, transmitter 0.15 m; camera–GPS rms 11.1 m out of sample

Antenna labels, one at a time. For each of the 23 labelled cameras, the camera's pose is refitted without its antenna label (from its horizon, tie points, GPS and priors) and the antenna is refitted from the other labels. The withheld ray then misses the antenna by a median of 1.04 m and at most 3.18 m (IMG_2232); 13 miss by more than 1 m. The antenna estimate itself is stable: over the 23 refits its jackknife standard error is (0.58, 0.36, 0.26) m in (E, N, U). The large misses come from poses that their own label partly sets: without its label, IMG_2232 moves 6.1 m.

Transmitter without the unconditioned pair. Withholding the transmitter labels of IMG_2210/2211 and refitting those two poses, the six transmitter-era cameras and the transmitter moves the six-camera transmitter by only 0.02 m. The pair's poses move 3.4–4.5 m, and their rays then miss the six-camera transmitter by 1.9–2.6 m and miss each other by 2.5 m; their closest point is +0.21 m in height and 2.6 m horizontally from it. The historical 1 m vertical disagreement does not return, but the pair cannot check the transmitter to better than about 2 m: their poses are not determined well enough without the label being tested.

GPS withheld. Removing every HEIC GPS term (horizontal fixes, the common bias and the EXIF altitude), the best rigid translation of the whole network is 0.15 m, and refitting every pose, the antenna and the transmitter moves the antenna 0.21 m and the transmitter 0.15 m. The horizons and landmarks against the DEM, not the

GPS, set the network's position. The camera-to-HEIC distances are then an out-of-sample check: rms 11.1 m (in sample, 11.0 m), worst 33.7 m, with a common offset of 3.6 m ± 2.0 m. Phone GPS therefore confirms the network's absolute horizontal placement only to a few metres; the DEM's own survey offset (provisionally zero) is not tested.

5 The joint MCMC

The v0004 run samples every usable camera pose, 3,288 landmark positions, the antenna, the transmitter, the common GPS bias and the two extra label scatters: 5 chains of 2000 retained sweeps over 213 global coordinates (§6). It did not converge, and nothing in it is quoted as a posterior. Figures 5 and 6 show its draws against the selected v0004 state. What they show:

- **v0004 is a state the chains do not return to.** The selected state's exact log density is −7366.92. The chains end between −7798 and −7487, and the best complete state any of them saved is 32 lower than the selected one (Fig. 6c). Their draws therefore sit beside the selected point, not around it (the last two rows of Table 1 give the offsets of the pooled chain means from the selected state and the spread of the chain means).
- **The sampler kernel is the bottleneck.** The median bulk effective sample size is 3237 for the GPS bias, which is drawn exactly by Gibbs, and 154 for the label scatters; it is 9, 10 and 11 for the camera positions, angles and focal lengths, 20 for the antenna and 6 for the transmitter.
- **Chains separate most where the tie-point network is weakest.** The ratio of between-chain to within-chain spread exceeds 1.5 for 14 of 29 cameras. It is largest for IMG_2199 and IMG_2159 (7.0 and 6.0), the two transmitter-era frames whose tie points link only each other, so the pair can slide together; then IMG_2203 (4.4, with no tie points). It also exceeds 1.5 for 11 of the 23 91 m cameras, each with at least 172 tie points. Tie points alone therefore do not prevent separate modes in this run: 13 of the 19 cameras with tie points separate, against 1 of the 10 without.
- **The focal–depth degeneracy.** For IMG_2199, the chain means lie 2–5 m from its labelled target with focal lengths 2970–3011 px, the farther chain means with the longer lens (rank correlation +0.9 over 5 chains).
- **The draws spread along each camera's line of sight** (Fig. 5a): for 11 of the 14 separated cameras the long axis of the pooled east/north draws lies within 30° of the line to the labelled target (median over all cameras 6°). The cameras slide toward and away from the scene rather than across it.

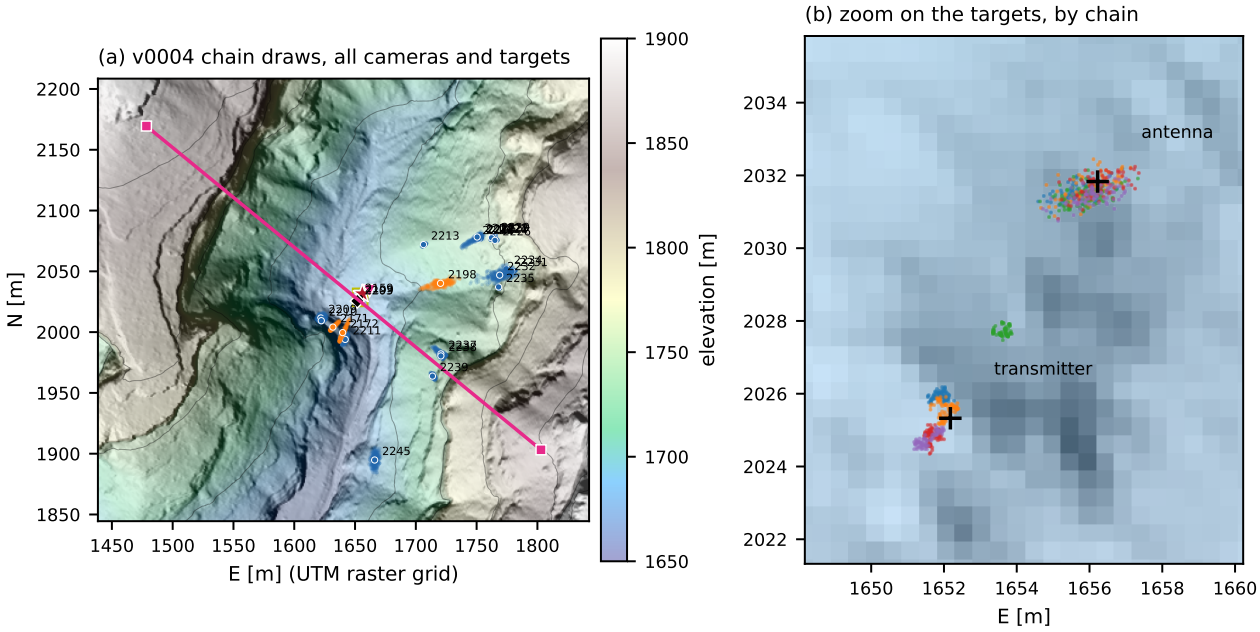


Figure 5: East/north draws of the five v0004 chains on the DEM hillshade (every 25th retained sweep). Elevation is coloured over the relief. (a) Every camera (blue: 91 m parallax set; orange: transmitter era; pooled over chains, labelled at its median), the antenna (red) and the transmitter (black), with the highline between its GPS anchors and the tiedown (pink); the yellow box is the zoom in (b). (b) The two targets, coloured by chain; crosses are the selected v0004 state. The draws need not centre on the crosses: v0004 is the highest-density state found, which no chain returns to (Fig. 6c).

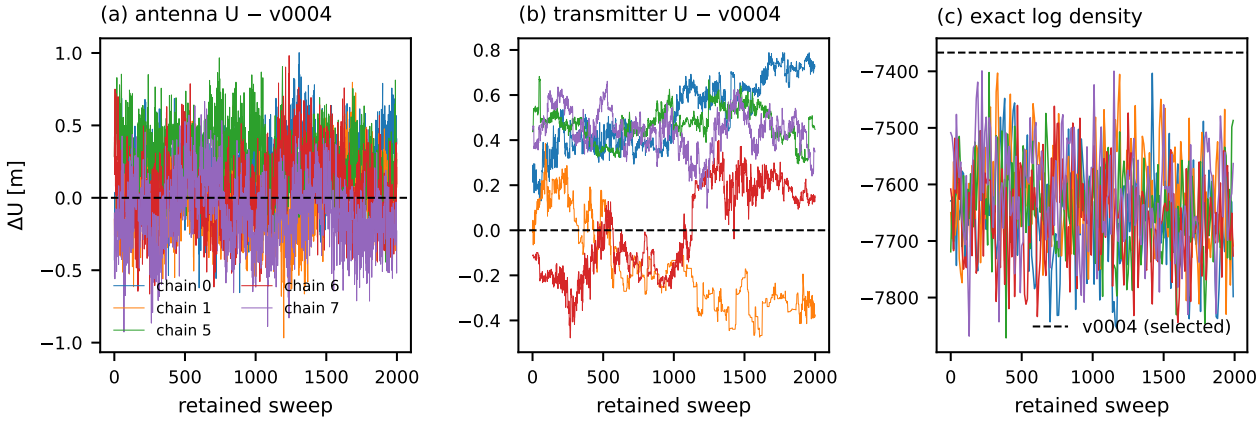


Figure 6: The five non-converged v0004 chains. Height of (a) the antenna and (b) the transmitter per chain, minus the selected v0004 value (dashed). (c) Exact log density per chain; dashed, the selected state's (−7366.92). The selected state is a higher-density point than any the chains reach, so their heights need not centre on zero.

6 Corrected model, proposal pilots and the v0004 fit

The October pilot ledger in Table 3 covers the original raw-f proposal, its log-f repair, the float32 DEM change, collective network moves, the loose and narrow focal-prior runs, the corrected EXIF run, paired height–northing, uniform angle directions, and state-aware angle banks. Each row gives the target, run size, and outcome. The old width-only EXIF conversion put the wrong prior centre on portrait photographs; it used the short image side in place of the diagonal. The corrected conversion uses the image diagonal divided by the 35-mm frame diagonal. The main and telephoto log-f prior width is 3%; the two 14-mm ultrawide frames (IMG_2198 and IMG_2203) retain 25% because their distortion model remains suspect. The focal-prior runs before this correction cannot establish lens behaviour.

Table 3: October camera proposal studies. All failed to establish a converged joint posterior.

pilot	target and retained scope	$\hat{R}$ fail	result
Original raw-f pilot	int32 DEM, width EXIF; 8 × (300+200)	212	Focal coordinates froze; raw-f proposal and log-f target measure disagreed.
Log-f repair pilot	int32 DEM, width EXIF; 8 × (300+200)	212	All cameras moved; transmitter and coupled geometry stayed separated.
Float32 paired collective pilot	float32 DEM, width EXIF; 2 starts × 2 arms, 200 retained	not assessed	One collective network direction reduced camera separation; neither arm converged.
Float32 five-chain run	float32 DEM, width EXIF; 5 × (300+2000)	not assessed	Transmitter position and two transmitter-era cameras remained separated.
Loose focal-prior five-chain run	float32 DEM, width EXIF; 5 × (300+2000)	213	No global passed; it exposed the width-only EXIF conversion error.
3% focal-prior five-chain run	float32 DEM, width EXIF; 5 × (300+2000)	209	Narrow prior tested the wrong focal centre; no lens-physics inference.
Corrected EXIF mixed-prior run	float32 DEM, diagonal EXIF; 5 × (300+2000)	212	Only GPS bias northing passed; height and orientations stayed separated.
Paired height/northing pilot	corrected target; 2 starts × 2 arms, 100+200	212	Height movement per CPU rose, but height-chain separation grew.
Uniform eight-direction pilot	corrected target; 2 starts × 2 arms, 100+200	211	Angle movement rose without effective cross-chain mixing.
State-aware bank pilot	corrected target; 2 starts × 2 arms, 100+200	211	Local bank use rose as intended; all-coordinate convergence still failed.
UTM-target five-chain run	DEM v0002, raw HEIC UTM GPS; 5 × (300+2000)	212	The carried v0003 point has higher exact density than every saved new-target draw; no convergence.

The full corrected run (`derived/geometry_posterior/v0003/full_20261006`) started five chains from the latest saved valid complete states. Each chain ran 300 warmup and 2,000 retained sweeps. It used the float32 DEM, corrected diagonal EXIF conversion, and mixed 3%/25% focal priors. Across 213 global coordinates, 210 fail $\hat{R} \leq 1.01$, 210 fail bulk ESS ≥ 400, and 199 fail tail ESS ≥ 400; the worst $\hat{R}$ is 2.87 (`cam2159_n`). The two retained halves also fail, which contradicts any claim that the chains reached a common stationary distribution. The selected complete state, chain 0 at retained sweep 810, has exact target log density -7367.73. It gives antenna (1656.22, 2031.83, 1778.60) and transmitter (1652.18, 2025.33, 1683.48) metres in the historical tangent working grid. This is the old-frame `v0003` fitted state, not a posterior mean or an uncertainty-bearing position. The previous corrected run retained a state with higher exact log density (-7366.74 versus -7367.73), so the v0003 point is the latest published old-frame update rather than evidence of improved optimization.

The UTM-target run `derived/geometry_posterior/v0004` used the same five chain starts, 300 warmup and 2,000 retained sweeps per chain, but re-extracted HEIC GPS positions onto DEM v0002. It kept the corrected diagonal EXIF conversion and mixed focal-prior widths. The old collective proposal directions were not reused for the changed target; the standard blocked camera, landmark and network-shift proposals were recalculated. The eigsep_terrain code ran at `3ba16e6`. The checksum-pinned input manifest and five-chain checkpoint audit are in that product. Of 213 globals, 212 fail $\hat{R} \leq 1.01$, 211 fail bulk ESS ≥ 400, and 200 fail tail ESS ≥ 400; the worst $\hat{R}$ is 3.11 (`cam2199_ph`). These chains do not support a posterior estimate.

The highest saved new-chain complete state has new-target log density -7399.23. The previously selected v0003 photo geometry, re-evaluated on the UTM target with its common GPS bias fitted conditionally, has higher density -7366.92. Geometry `v0004` therefore keeps those camera, landmark, antenna and transmitter coordinates on the UTM raster grid, with common GPS bias (+1.76, +3.11) m (E,N). It is the adopted release (§4) because it is the highest-density valid state on the corrected target, not because any chain reached it; it is a point selection, not a convergence result or a physical survey-offset estimate. For the 27 modeled HEIC photographs with GPS, camera-to-fix horizontal RMS changes from 22.1 m under the historical mapping to 11.0 m in UTM; 26 fixes move closer and the largest remaining separation is 33.6 m (IMG_2215). Both camera eras improve separately. This comparison is in-sample because HEIC GPS entered the fit; the GPS-withheld refit in §4.1 gives the out-of-sample comparison.

The study record is `derived/geometry_memo_inputs/v0001/pilot_review.json` and its source was executed at legacy terrain commit `85cd7af`; its pilot inputs and code checksums are recorded there. The v0003 run report and manifest pin its chain files, model source commit `85cd7af`, starting states and selected complete state. The v0004 manifest additionally pins its UTM DEM, re-extracted HEIC GPS, eigsep_terrain commit, chain inputs and selected state. This memo generates the displayed pilot table and both version summaries from those records and rejects changed input checksums.

7 Azimuth zero and the LIDAR check

Box-air's azimuth is referenced to the potentiometer zero, at which the dipole arm lies along the highline (memo 001 §7). The highline's two ground anchors were measured with a GPS receiver, and their bearing, 307.84° east to west over 420 m, is immune to any common GPS offset. The boresight plane at zero azimuth is perpendicular to it: **az-zero = 37.84° true**, ±0.6° for ±3 m per anchor. With the anchors placed on the UTM grid and the antenna at v0004, the antenna hangs 6.37 m off the straight anchor-to-anchor line (station 195.0 m of 420); the GPS-measured pulley-plate tiedown is 11.75 m off it on the same side (station 192.0 m), consistent with the tethers pulling the pulley plate off the span. These offsets depend on the provisional zero horizontal survey offset; the historical tangent frame, with its (−11, 36) m offset, put the antenna 10 m off the line. So the anchors measure a bearing, not the antenna's position.

The platform LIDAR gives an independent check of both the antenna position and az-zero (Fig. 7; Tables 5 and 4; `derived/lidar_constraint/v0002`). It is mounted 90° from the boresight, so as elevation sweeps it ranges the canyon walls. Its rays are marched through the DEM from the v0004 antenna, with the full mount rotation; the bearings (az-zero and the highline axis) are true bearings, applied on the raster as grid bearings, which differ by 1.52°. With az-zero fixed at the GPS value, the sweep returns fit with a median absolute residual of 1.21 m (147 of 151 rays hit terrain; 27 are more than 10 m from the model or miss), best at an antenna-height offset of +0.00 m. Leaving az-zero free moves it to 36.27° ± 0.45° (bootstrap over the sweep rays), 1.57° ± 0.75° from the GPS value, a 2.1σ tension. The long near-nadir dwell (488 returns) measures 92.31 m against 94.50 m predicted, a -2.2 m residual; the LIDAR's lever arm from the labelled antenna point is not modelled.

Changing one thing at a time (Table 4) separates the two changes from the historical fit, which applied true bearings directly on the raster at the v0001 antenna and gave 36.76° (1.4σ). Moving only to the v0004 antenna lowers the tension to 1.1σ; correcting only the frame raises it to 2.1σ. On the UTM grid the sweep is best at dU = +1.00 m for the v0001 antenna and +0.00 m for v0004, so the LIDAR favours the v0004 height. The tension at v0004 comes from the frame correction and is not yet explained: the DEM's horizontal survey offset is provisionally zero, and an offset of the terrain relative to the GPS anchors would rotate the LIDAR fit.

Table 4: The LIDAR terrain fit with the frame and the antenna changed one at a time. "Tangent" applies true bearings directly on the raster, as the historical fit did; "UTM grid" converts them to grid bearings. dU is the best antenna-height offset with az-zero at the GPS value; the tension is (GPS − free)/$\sqrt{0.6^2 + \sigma_{\mathrm{boot}}^2}$. The last row is the adopted check.

frame	antenna	dU [m]	MAD [m]	free az-zero [deg]	GPS − free [deg]	tension [σ]	dwell res. [m]
tangent (as v0001)	v0001	+0.25	1.15	36.76 ± 0.44	+1.07	1.4	-1.2
UTM grid	v0001	+1.00	1.28	36.32 ± 0.41	+1.51	2.1	-2.2
tangent (as v0001)	v0004	-0.75	1.16	36.89 ± 0.58	+0.94	1.1	-1.2
UTM grid	v0004	+0.00	1.21	36.27 ± 0.45	+1.57	2.1	-2.2

Table 5: Azimuth zero (the boresight-plane bearing at potentiometer zero).

route	az-zero [deg]	uncertainty	note
Highline GPS anchors	37.84	±0.6 (±3 m per anchor over 420 m)	bearing between two surveyed points; immune to common GPS offset
LIDAR terrain fit, az-zero free	36.27	2.09σ from GPS	MAD 0.92 m; fixing it at the GPS value costs 0.29 m
LIDAR terrain fit, az-zero fixed	37.84	fixed	MAD 1.21 m; 147 of 151 sweep rays hit terrain

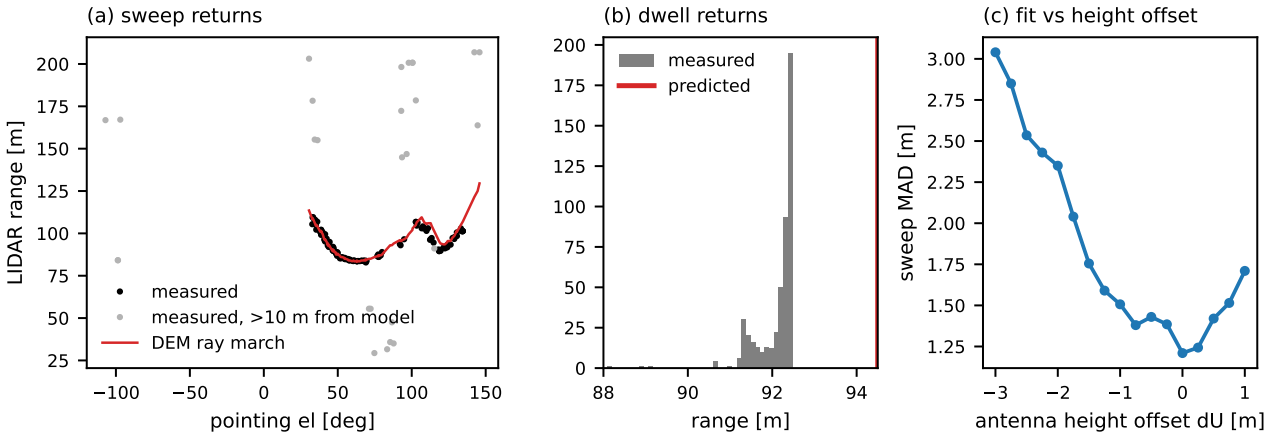


Figure 7: LIDAR constraint at the v0004 antenna on the UTM grid, with az-zero fixed at the GPS value. (a) Sweep returns against the DEM ray march. (b) Near-nadir dwell returns and the predicted range. (c) Sweep fit quality against an antenna-height offset.

8 Heights

The 91 m-era antenna is 94.1 m above the DEM surface at its v0004 (E, N) (the historical release gave 93.4 m, as quoted in memo 001, Table 4); the LIDAR dwell reads 92.3 m, a range to whatever ground it struck; the field notes' 91 m has no recorded datum. The only height with a stated datum, 87.5 m, is "from the tether anchor", whose position is not recorded. These are different quantities and must not be averaged (`curation/height_references.json`).

9 Transmitter antenna orientation

The transmitter antenna is a pair of crossed dipoles on a square feed; alternate comb teeth come from alternate dipoles, rx6 (arm 0) and rx1 (arm 1) (memo 001 §2). Its orientation is measured from IMG_2203, a transmitter-era frame taken about 2 m above the feed, looking down on it (Fig. 8). The four outer corners of the feed face, picked by Aaron, are fitted as a planar square seen through the camera's fitted pose, with its perspective and radial distortion and no assumed size. The rx6 axis joins the two quadrants marked "6" in the field; rx1 joins the other two.

Bearings here are true bearings. The poses live on the UTM raster grid, so an axis computed on it is a grid bearing; true = grid − 1.52° at the transmitter. At the v0004 pose of IMG_2203 the square fits with a corner rms of 1.7 px and a tilt of 1.3° from horizontal, and **the rx6 axis lies at 165.5°, rx1 at 75.5°** (mod 180). A second planar solution, nearly vertical, fits worse and is rejected because the feed sits flat in its dish. IMG_2203's pose is uncertain: it is transmitter-conditioned and has no tie points. Across 500 draws of the v0004 chains the rx6 axis is 165.5° with sd 0.74°; the chains did not converge (R-hat up to 2.6 for this camera), so that spread is not a confidence interval. Moving the corners by 2 px, about their rms, spreads it by 0.39°, and letting the face be 10% off square moves it over 165.3–165.8°. Against the historical solution (`tx_feed_orientation/v0001`, at the historical pose and MCMC draws), the axis moves −0.28° on the grid; the larger change is the frame: the historical 167.3° was a grid bearing reported as true.

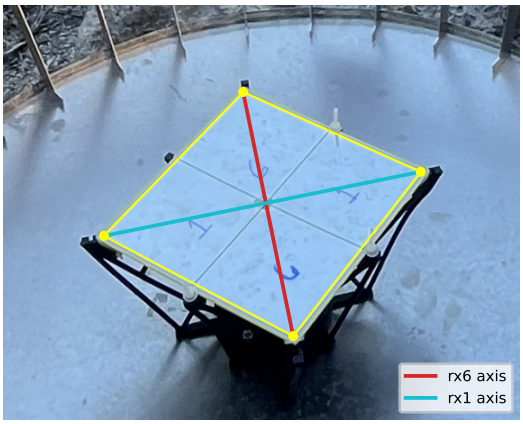
Adopted: rx6 axis 165.5° ± 1° true (NNW–SSE), rx1 axis 76° (ENE–WSW).

The theodolite-app reading in IMG_2204, corrected to the arm as in the legacy geometry memo, is 166.71°, 1.2° from this axis. Its north reference (true or magnetic-corrected) is not recorded, so the comparison is good only to a few degrees. IMG_2202 agrees once it is read correctly. It was taken at the transmitter, looking across the feed, and its horizon fit gives a camera heading of 131.3° true; an independent match of 20 canyon features gives 131.9°, so the heading is sound. The legacy geometry memo read that heading as a polarization axis, which it is not: the clear zip ties that attach the feed's corners to its support sit to the left and right of the sightline, so the camera looks across a flat edge of the square, about 45° from both arms. The feed also sits left of the image centre, so the camera to feed bearing is about 124–125°. That places the arms near 79–80° and 169–170°, within 4–5° of the IMG_2203 measurement, as expected for a sighting by eye. Comparing this with the polarization angle fitted in the beam maps belongs to the beam memo. The analysis is `scripts/marjum-2026-07/geometry/tx_feed_orientation.py` in `data-analysis` (commit 03f3219), which reproduces the historical product from its frozen inputs; its output is `marjum-2026-07/derived/tx_feed_orientation/v0002` (Appendix C).

10 Open questions

- Poses that their own labels set: without its antenna label IMG_2232 moves 6.1 m, and IMG_2210/2211 move 3.4–4.5 m without their transmitter labels. These set the bounds; better-constrained poses (more tie points, a converged joint posterior) would tighten them.
- The antenna-label misfit: 10 of 23 antenna rays miss v0004 by more than 1 m, and the fitted extra label scatter is 27 px.
- The 2.1σ tension between the LIDAR and GPS az-zero on the UTM grid, and the DEM's untested horizontal survey offset.
- A converged joint posterior: the tested focal, collective, paired and state-aware proposals did not mix. The ultrawide distortion model and the zero-tie-point transmitter cameras remain concrete sources of weak geometry.

(a) IMG_2203: feed corners (Aaron)



(b) rx6 axis across the v0004 chain draws of IMG_2203

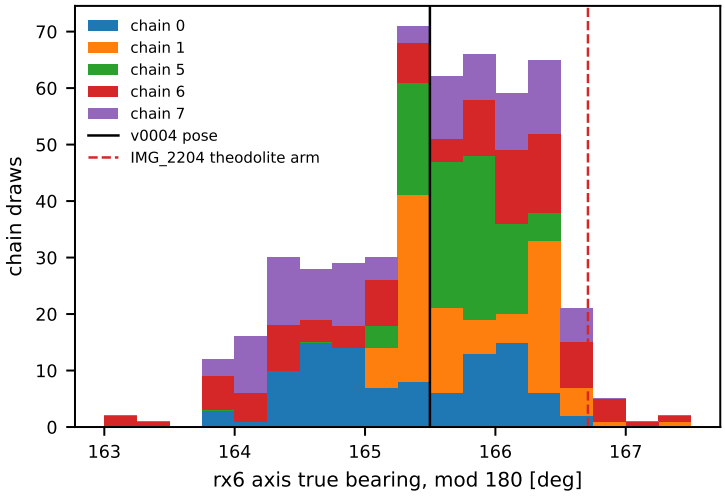


Figure 8: (a) The transmitter feed in IMG_2203 with Aaron's corner picks; the rx6 axis joins the quadrants marked "6", rx1 the other two. (b) The rx6 axis bearing solved through each of 500 draws of the v0004 chains for IMG_2203, stacked by chain, as a true bearing; the solid line is the v0004 release pose, dashed the IMG_2204 theodolite-app arm (north reference not recorded).

- The north reference of the IMG_2204 theodolite-app reading, needed to use it as a check on the feed axis at better than a few degrees.
- The tether anchor's position, needed to put the field heights on a datum.
- Whether the antenna held its horizontal position across lifts (no parallax frames outside the 91 m era).

Revision history

2026-10-10 The MCMC section, its figures and the chain rows of Table 1 now come from the v0004 chains, not the historical run, which had been plotted against v0004. They show v0004 to be a higher-density state than any chain reaches, and they overturn the historical finding that cameras without tie points split into separate modes: most separating cameras have tie points. The feed orientation is re-solved at the v0004 pose and draws and given as a true bearing (165.5°); the earlier 167.3° was a grid bearing.

2026-10-09 Held-out refits and a UTM-grid LIDAR fit give v0004 bounds of 3.2 m (antenna) and 2.7 m (transmitter); the LIDAR–GPS az-zero tension rises to 2.1σ once the LIDAR bearings are applied on the grid.

2026-10-08 Adopted geometry `v0004` as the release: positions, figures, ray misses and horizon checks are now at the v0004 poses on the UTM raster grid, with no uncertainty established; the `v0001` positions and bounds are historical.

2026-10-07 Corrected the historical tangent-versus-UTM frame description and added the v0004 UTM-target run, its nonconvergence, the carried point-fit selection and the HEIC GPS comparison.

2026-10-07 Moved the October camera proposal study into this memo, distinguished the corrected v0003 provisional point fit from a posterior, and migrated the memo's historical terrain inputs into a versioned campaign product.

2026-09-30 First version, graduating the MCMC review, geometry explorer and LIDAR explorer notebooks (`data-analysis/notebooks/arp/marjum-2026-07`) and the geometry parts of legacy MEMO-012. Same day: the transmitter antenna orientation (§9), withdrawing MEMO-012's reading of IMG_2202's camera heading as a polarization axis; and per-image solutions (Appendix B).

A Image inventory

From release `v0004` (`imgs/fits/v0004_marjum_geometry/cameras.jsonl`), which carries the geometry and labels the v0004 model used (its IMG_2211 transmitter label is a re-pick about 4 px from the v0001 one). EXIF times from the photographs. "TX-cond.": fitted with the transmitter label and no independent tie to the antenna; labels: A antenna, T transmitter.

image	EXIF (MDT)	group	pose	lens	f [px]	labels	exclusion reason
IMG_2159	07-15 16:49:12	tx-era	usable (TX-cond.)	0	2979	T	
IMG_2171	07-15 16:57:26	tx-era	usable (TX-cond.)	0	4442	T	
IMG_2172	07-15 16:58:00	tx-era	usable (TX-cond.)	0	4533	T	
IMG_2198	07-17 09:32:34	tx-era	usable (TX-cond.)	1	1541	T	
IMG_2199	07-17 09:36:14	tx-era	usable (TX-cond.)	0	2993	T	
IMG_2203	07-17 09:05:34	tx-era	usable (TX-cond.)	1	1618	T	
IMG_2209	07-17 14:42:18	91 m parallax	usable	1	2322	A	
IMG_2210	07-17 14:42:33	91 m parallax	usable	1	2698	A+T	
IMG_2211	07-17 14:43:29	91 m parallax	usable	1	2444	A+T	
IMG_2213	07-17 14:47:30	91 m parallax	usable	0	3187	A	
IMG_2214	07-17 14:49:05	91 m parallax	usable	0	6552	A	
IMG_2215	07-17 14:49:07	91 m parallax	usable	0	6405	A	
IMG_2216	07-17 14:49:14	91 m parallax	usable	1	2554	A	
IMG_2217	07-17 14:49:22	91 m parallax	usable	0	3905	A	
IMG_2218	07-17 14:49:28	91 m parallax	usable	0	8307	A	
IMG_2219	07-17 14:49:40	91 m parallax	excluded	—	—	A	Borderline registration: 67.9 px raster-horizon RMS and sky NLL 9.56; not promoted
IMG_2220	07-17 14:50:48	91 m parallax	usable	0	5414	A	
IMG_2221	07-17 14:51:05	91 m parallax	usable	0	3338	A	
IMG_2222	07-17 14:51:11	91 m parallax	usable	0	4626	A	
IMG_2223	07-17 14:51:13	91 m parallax	usable	0	4644	A	
IMG_2224	07-17 14:51:22	91 m parallax	usable	0	6586	A	
IMG_2225	07-17 14:52:02	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2226	07-17 14:52:14	91 m parallax	usable	0	3866	A	
IMG_2227	07-17 14:52:44	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2228	07-17 14:52:49	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2230	07-17 14:56:02	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2231	07-17 14:56:47	91 m parallax	usable	0	7799	A	
IMG_2232	07-17 14:57:03	91 m parallax	usable	1	1894	A	
IMG_2234	07-17 14:58:12	91 m parallax	usable	0	3788	A	
IMG_2235	07-17 14:59:46	91 m parallax	usable	1	2936	A	
IMG_2237	07-17 15:03:21	91 m parallax	usable	0	8989	A	
IMG_2238	07-17 15:03:43	91 m parallax	usable	1	2888	A	
IMG_2239	07-17 15:04:30	91 m parallax	usable	0	3652	A	
IMG_2241	07-17 15:05:22	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2242	07-17 15:06:34	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum

image	EXIF (MDT)	group	pose	lens	f [px]	labels	exclusion reason
IMG_2243	07-17 15:07:25	91 m parallax	excluded	—	—	A	Rejected after registration entered a false terrain minimum
IMG_2245	07-17 15:10:03	91 m parallax	usable	0	5362	A	

B Per-image solutions

One panel per usable image, at the v0004 poses. Green: the sky/ground boundary segmented from the photograph (`derived/geometry_memo_inputs/v0001/img_seg_IMG_*.npz`); red: the DEM horizon projected through the fitted pose. Plus signs are the labelled antenna (yellow) and transmitter (orange) pixels; crosses are the v0004 positions projected through the pose. Each title gives the median horizon offset and, for each label, the offset in pixels and the perpendicular miss of the labelled ray in metres. The table repeats these, with the distance to the antenna. The "horizon" offset is the median over image columns where both boundaries exist and the ground below the segmented boundary is not tree (trees hide the terrain skyline). Pixel offsets scale inversely with range: the transmitter-era cameras were 2–30 m from the transmitter, so a 0.5 m miss can be hundreds of pixels.

				antenna			transmitter	
image	group	f [px]	horizon [px]	range [m]	miss [px]	miss [m]	miss [px]	miss [m]
IMG_2159	tx-era	2979	9	—	—	—	13	0.02
IMG_2171	tx-era	4442	9	—	—	—	10	0.06
IMG_2172	tx-era	4533	5	—	—	—	12	0.08
IMG_2198	tx-era	1541	8	—	—	—	12	0.42
IMG_2199	tx-era	2993	9	—	—	—	14	0.01
IMG_2203	tx-era	1618	17	—	—	—	9	0.01
IMG_2209	91 m parallax	2322	5	102	33	1.28	—	—
IMG_2210	91 m parallax	2698	5	106	71	2.24	6	0.07
IMG_2211	91 m parallax	2444	9	106	41	1.24	14	0.17
IMG_2213	91 m parallax	3187	8	85	19	0.45	—	—
IMG_2214	91 m parallax	6552	10	111	29	0.48	—	—
IMG_2215	91 m parallax	6405	10	107	63	1.03	—	—
IMG_2216	91 m parallax	2554	5	111	10	0.41	—	—
IMG_2217	91 m parallax	3905	12	111	45	1.05	—	—
IMG_2218	91 m parallax	8307	15	109	31	0.41	—	—
IMG_2220	91 m parallax	5414	12	120	17	0.36	—	—
IMG_2221	91 m parallax	3338	6	118	21	0.68	—	—
IMG_2222	91 m parallax	4626	8	118	11	0.26	—	—
IMG_2223	91 m parallax	4644	11	119	17	0.39	—	—
IMG_2224	91 m parallax	6586	15	117	42	0.74	—	—
IMG_2226	91 m parallax	3866	11	119	15	0.41	—	—
IMG_2231	91 m parallax	7799	19	124	77	1.18	—	—
IMG_2232	91 m parallax	1894	8	113	21	1.22	—	—
IMG_2234	91 m parallax	3788	9	122	25	0.78	—	—
IMG_2235	91 m parallax	2936	8	118	87	2.93	—	—
IMG_2237	91 m parallax	8989	10	92	69	0.69	—	—
IMG_2238	91 m parallax	2888	6	95	70	2.01	—	—
IMG_2239	91 m parallax	3652	7	103	96	2.48	—	—
IMG_2245	91 m parallax	5362	6	163	16	0.47	—	—

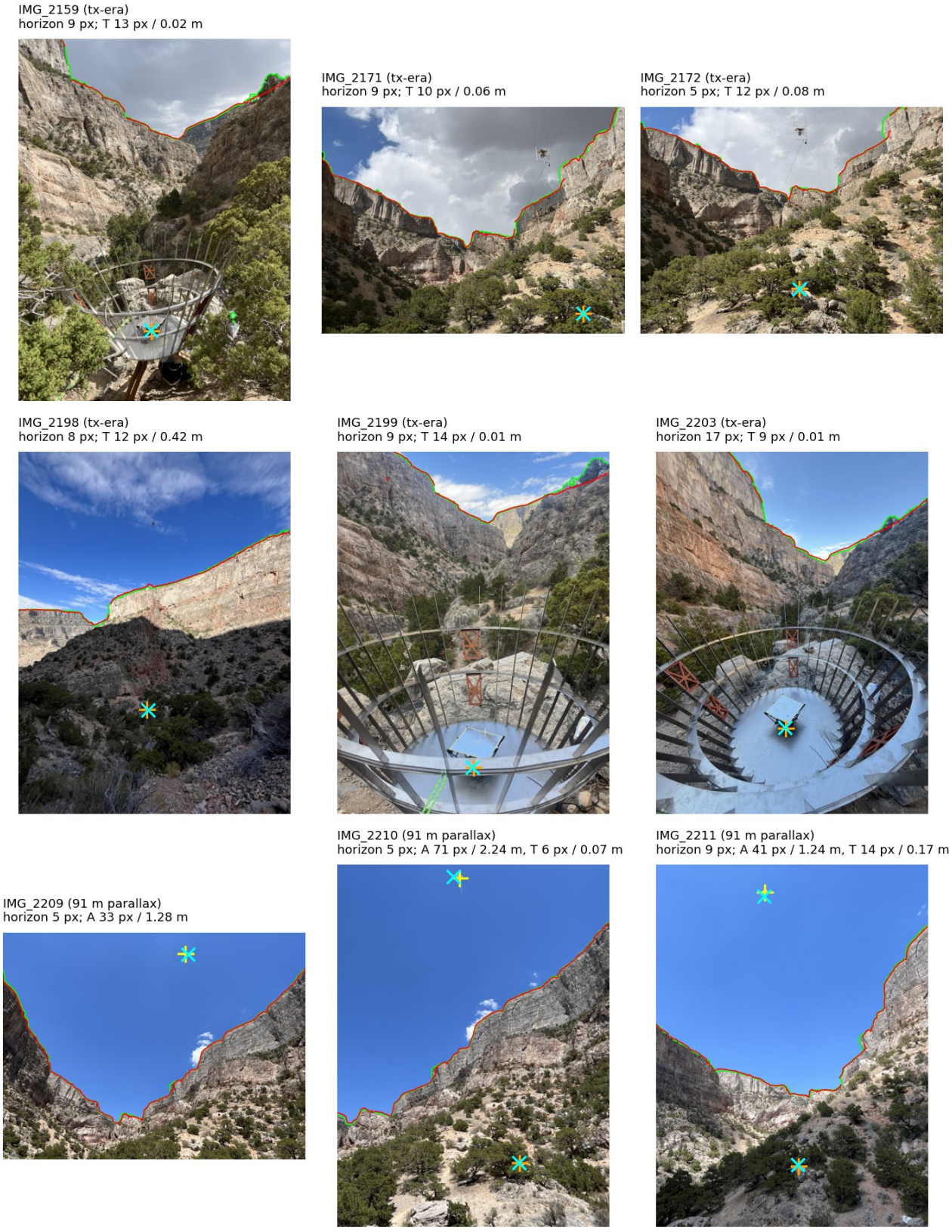


Figure 9: Per-image solutions, page 1 of 4; see Appendix B.



Figure 10: Per-image solutions, page 2 of 4; see Appendix B.

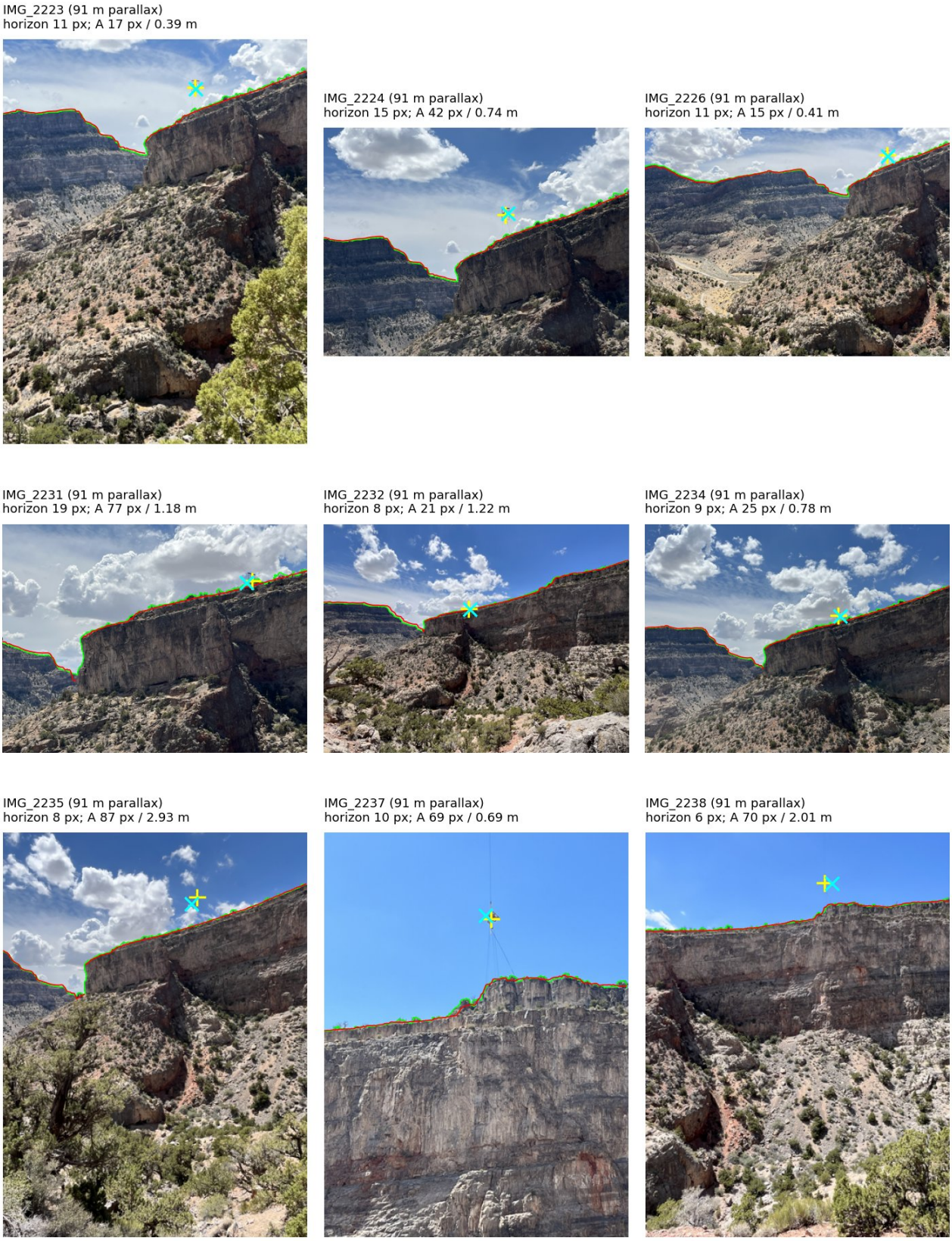


Figure 11: Per-image solutions, page 3 of 4; see Appendix B.

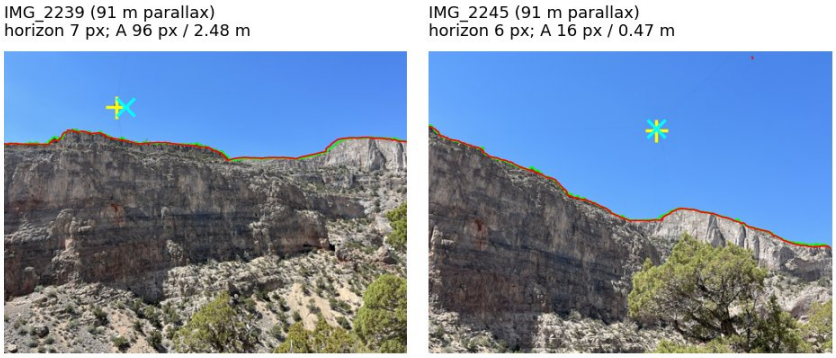


Figure 12: Per-image solutions, page 4 of 4; see Appendix B.

C Reproducing this memo

From this directory, with the project environment:

```
python make_products.py
pdflatex memo-002.tex && pdflatex memo-002.tex
```

Inputs, with the first 16 hex digits of their SHA-256. The DEM, the geometry, the LIDAR constraint and the transmitter-feed orientation are products under `marjum-2026-07/derived/`; each has a manifest recording its code and inputs. The v0004 model inputs (labels, distortion and tie-point observations) are read from the product's copy in `derived/geometry_posterior/v0004/inputs/model_v0002`, pinned by checksum here and in its `copied_inputs.json`. The v0004 model sources are pinned in its `input_manifest.json`.

input	SHA-256 (first 16)
marjum-2026-07/imgs/fits/v0004_marjum_geometry/cameras.jsonl	f9d99398d18686f4
marjum-2026-07/imgs/fits/v0004_marjum_geometry/shared.json	8fcf5360cbfa14a0
marjum-2026-07/imgs/fits/v0004_marjum_geometry/manifest.json	1fef0c9983a98b40
marjum-2026-07/imgs/fits/v0001_marjum_geometry/cameras.jsonl	2ed744f833f689e9
marjum-2026-07/imgs/fits/v0001_marjum_geometry/shared.json	234c1d5ab7142318
marjum-2026-07/derived/geometry_memo_inputs/v0001/pilot_review.json	18ac3ba300391bda
marjum-2026-07/derived/geometry_memo_inputs/v0001/manifest.json	758f80c9fc7e2f06
marjum-2026-07/derived/geometry_posterior/v0003/manifest.json	605a8d216491d23f
marjum-2026-07/derived/geometry_posterior/v0003/full_20261006/convergence_report.json	4f9762fbe87cd130
marjum-2026-07/derived/geometry_posterior/v0003/provisional_geometry.npz	06024c369cde40dc
marjum-2026-07/derived/geometry_posterior/v0004/manifest.json	29a747c8b9fc5ce9
marjum-2026-07/derived/geometry_posterior/v0004/run/convergence_report.json	fba0fbdee321e1cd
marjum-2026-07/derived/geometry_posterior/v0004/run/chain_0.npz	b9096d471e805897
marjum-2026-07/derived/geometry_posterior/v0004/run/chain_1.npz	01ec65114df3ab40
marjum-2026-07/derived/geometry_posterior/v0004/run/chain_5.npz	caa8384549e6f341
marjum-2026-07/derived/geometry_posterior/v0004/run/chain_6.npz	9380a8d628693c81
marjum-2026-07/derived/geometry_posterior/v0004/run/chain_7.npz	e012fd1f0ba30975
marjum-2026-07/derived/geometry_posterior/v0004/gps_comparison.json	53ffd1d056b902b2
marjum-2026-07/derived/geometry_posterior/v0004/provisional_geometry.npz	6e611506298f05aa
marjum-2026-07/derived/geometry_posterior/v0004/inputs/model_v0002/fit_transmitter.npz	5ae3909a7b42abdd
marjum-2026-07/derived/geometry_posterior/v0004/inputs/model_v0002/meta.json	4c0e5de2e0f07d0f
marjum-2026-07/derived/dem/v0002/marjum_dem.npz	e890231c970be4ed
marjum-2026-07/derived/lidar_constraint/v0002/results.npz	69b3b59a81855b45
marjum-2026-07/derived/lidar_constraint/v0002/summary.json	d684aa673f7c0d05
marjum-2026-07/derived/lidar_constraint/v0002/manifest.json	facb7402cabe972e
marjum-2026-07/derived/geometry_checks/v0001/manifest.json	7b30aeac3de211c6
marjum-2026-07/derived/tx_feed_orientation/v0002/summary.json	1af458e39f88f320
marjum-2026-07/derived/tx_feed_orientation/v0002/draw_diagnostics.json	610cae8dbc7fe160
marjum-2026-07/derived/tx_feed_orientation/v0002/manifest.json	9b7a23f5e11ebe7a
marjum-2026-07/derived/tx_feed_orientation/v0001/inputs/tx_feed_picks.json	d5f3fc6a5b4891b3
eigsep (memo, campaign) commit	f1dc910
legacy analysis commit	8a09806
generated	2026-10-10 05:17 UTC